Accelerated Filtering on Graphs using Lanczos method

Scientific Computing project report

Alberto Defendi

Abstract

We study Krylov methods for the fast evaluation of graph filters of the form $g(\mathcal{L})s$, where $\mathcal{L}$ is the (symmetric) graph Laplacian in the context of graph signal processing. We implement Lanczos and Arnoldi algorithm and we compare different orthogonalization strategies, together with a convergence-based stopping rule. We first discuss the computational costs of these variants. Then, through the experiments on Erdős-Rényi graphs we relate the iteration error to the true error. Finally, we report how runtime scales with graph size and sparsity.

Contents

1 Introduction

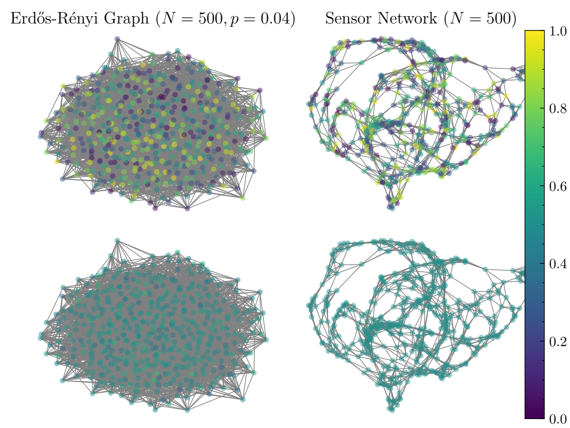
We introduce basic graph theory concepts and we recall some theoretical results used troughout the report.

1.1 Signal processing on graphs

We consider a weighted undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{W})$, where $\mathcal{V} \subset \mathbb{R}^N$ is the set of vertices, $\mathcal{E} \subset \mathbb{R}^M$ is the set of edges, and $\mathcal{W} : \mathcal{V} \times \mathcal{V} \to \mathbb{R}$ is a weight function. We represent the weight function with a $N \times N$ matrix W such that $W_{i,j} = \begin{cases} W(v_i, v_j), & \text{if } (v_i, v_j) \in \mathcal{E} \\ 0, & \text{otherwise} \end{cases}$ for all $i, j = 1, \ldots, |\mathcal{V}|$, and for our needs, we assume W to be symmetric, that is $W_{i,j} = W_{j,i}$.

In the study of graph signal processing, we model the idea of sending a signal to a node as assigning a value to a vertex with a function $s : \mathcal{V} \to \mathbb{R}$, whose values can be represented by a vector $s = s(\mathcal{V}) = [s_1, \ldots, s_N] \in \mathbb{R}^N$ where each entry $s_i := s(v_i)$ represents the signal sent over a node $v_i \in \mathcal{V}$. We also keep track of the weight of each vertex with a function $d(i) := \sum_{j=1}^N W_{i,j}$, and we call $D = \text{diag}(d(1), \ldots, d(N))$ the degree matrix.

Figure 1: Filtering signal on graphs.



In this setting, we introduce the graph Laplacian $\mathcal{L} = D - W$. By construction, $\mathcal{L}^T = \mathcal{L}$, thus the Lanczos method can be safely applied to our problem. This matrix has some other useful properties that we will list in the following Proposition.

Proposition 1.1. *The Laplacian matrix $\mathcal{L} \in \mathbb{R}^{N \times N}$ satisfies the following properties:*

1. *$\mathcal{L}$ is symmetric and positive semi-definite.*
2. *The smallest eigenvalue of $\mathcal{L}$ is 0 with corresponding eigenvector the constant one vector $\mathbf{1}$.*
3. *$\mathcal{L}$ has N non-negative real-valued eigenvalues $0 = \lambda_0 \leq \lambda_1 \leq \ldots \leq \lambda_{N-1}$.*

Proof. See [1]. □

Remark 1.2. In light of this proposition and by the spectral theorem, the Laplacian can be decomposed as:

$$\mathcal{L} = U \Lambda U^*,$$

where $U = [u_0, \ldots, u_{N-1}]$ is a $N \times N$ orthogonal matrix and we call it the *Fourier basis*, and $\Lambda = \text{diag}(\lambda_0, \ldots, \lambda_{N-1})$ is the matrix of its eigenvalues, which are monotonically increasing, and the vectors in U are ordered according to the eigenvalues.

Definition 1.3 (Graph filter)**.** A graph filter is a continuous function $g : \mathbb{R}^+ \to \mathbb{R}$. For our needs, we will assume g to be defined on the spectrum of $\mathcal{L}$.

Diagonalising the Laplacian would give an easy way to compute the function $g(\mathcal{L})$ by evaluating it on the eigenvalues of $\mathcal{L}$, which means computing $g(\mathcal{L}) = Ug(\Lambda)U^*$. In matrix notation, applying a filter to a graph $\mathcal{G}$ corresponds to the operation

$$s' := g(\mathcal{L})s = Ug(\mathcal{L})U^*s.$$

However, computing the Fourier basis for $\mathcal{L}$ is computationally expensive for large graphs, as it requires the full eigendecomposition of $\mathcal{L}$. This motivates the idea for using the Lanczos method for evaluating the matrix function $g(\mathcal{L})$.

We consider Erdős-Rényi and Sensor graphs, following PyGSP's [2] definitions.

Definition 1.4 (Erdős-Rényi graph)**.** We define the Erdős-Rényi graph $\mathcal{G}(N, p)$ with N nodes, and probability p, as the graph constructed by randomly connecting two nodes independently of any other edge, with probability p.

Definition 1.5 (Sensor graph)**.** A sensor graph $\mathcal{G}(N)$ is the graph constructed by picking N points on the square $[0, 1] \times [0, 1]$ and connecting points with edges using k nearest neighbours.

Remark 1.6. As stated in PyGSP sensor graph source code, the definition of Sensor graph changed in 2019, switching to generation via the k-means algorithm. This might give slightly different results from the 2015's article [3] we are trying to reproduce, but we don not investigate this further.

1.2 The Lanczos method

Definition 1.7 (Krylov subspace)**.** Given a matrix $A \in \mathbb{R}^{N\times N}$ and a vector $b \in R^N$, the Krylov subspace of order l is defined as the set $\mathcal{K}_l(A, b) = \text{Span}\left(b, Ab, A^2b, \dots, A^{l-1}b\right)$. We represent the basis of this subspace with a matrix $V_M = [v_1, \dots, v_M] \in \mathbb{R}^{N\times M}$.

From the Arnoldi relation,

$$AV_l = V_lH_l + h_{l+1,l}v_{l+1}e_l^*, \qquad H_l = V_l^*AV_l = \begin{bmatrix} h_{11} & \cdots & \cdots & h_{1,l} \\ h_{21} & h_{22} & & \vdots \\ & \ddots & \ddots & \vdots \\ & & h_{l,l-1} & h_{l,l}. \end{bmatrix},$$

we can derive the Lanczos method for a symmetric matrix A. Setting

$$\alpha_l := h_{l,l}, \qquad \beta_l := h_{l+1,l}.$$

From our hypothesis on A, it follows that the matrix H_l, which is both upper and lower Hessenberg matrix, is also symmetric; therefore it is tridiagonal—thus we rename it T_l. Hence the Arnoldi relation takes the form:

$$AV_l = V_l T_l + \beta_l v_{l+1} e_l^T, \qquad T_l = V_l^\top A V_l = \begin{bmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \ddots & \ddots & & \\ & \ddots & \ddots & \beta_{l-1} \\ & & \beta_{l-1} & \alpha_l \end{bmatrix}.$$

.

We now introduce a useful approximation for evaluating a matrix function using the Arnoldi projection.

Proposition 1.8. *Consider the filter $g : [0, \lambda_{max}] \to \mathbb{R}$ and a signal vector $s \in \mathbb{R}^N$. Then the Lanczos method approximates $g(\mathcal{L})s$ with:* [1]

$$g(\mathcal{L})s \approx V_M \|s\|_2 \, g(T_M) e_1 =: g_M. \tag{1}$$

Numerically, we often compute the eigendecomposition (Λ, U) from T_M, and then substitute $g(T_M) = Ug(\Lambda)U^\top$ in the equation above.

1.3 Algorithm

We begin by defining quantities that will be useful in describing our implementation of the algorithms.

Definition 1.9 (Errors)**.** We define the *iteration error* as $\|g_{M+j} - g_M\|_2$, where j is small, and the *true error* as $\|e_M\|_2 = \|g(\mathcal{L})s - g_M\|_2$.

Definition 1.10 (Error bounds)**.** We indicate with $\tau =: \tau_{\text{STOP}}$ the algorithm stopping threshold for the inequality $\|g_{M+3} - g_M\|_2 < \tau$, and with $\delta =: \delta_{\text{ORTH}}$ the orthogonality threshold, i.e when $\|V^\top V - I\|_2 < \delta$.

In floating-point arithmetic, the basis V_M can loose orthogonality, and orthogonalizing the incoming vector w only against v_j, v_{j-1} is insufficient, for example if the input matrix A is badly conditioned. Hence, the need to reorthogonalize the new vector w against all the previous basis vectors using some steps of the Gram-Schmidt process. To do so, we implement three algorithms, which we refer to as Lanczos (Algorithm 1) and Arnoldi (Algorithm 5), which we implement following [4] and [3]. These two algorithms are often called by the orthogonalization selection in Algorithm 4, and include a stopping criterion based on matrix function convergence. Before describing the algorithms, we explain how g_M is evaluated in the Lanczos process using Equation 1. Recall that $g_M := \|s\|_2 V_M g(T_M) e_1$, where $g(T_M)$ is computed by diagonalizing T_M: because T_M is symmetric, real and tridiagonal, computing its eigendecomposition is stable and efficient; indeed the structure of T_M allows us to use a divide and conquer algorithm for eigenvalues. We choose to represent it in banded form (isolating α_M and β_M) and using `scipy.linalg.eigh_tridiagonal` routine from Scipy, which costs $O(N^2)$.

[1]This result holds in general for any function $f : \Omega \to \mathbb{C}$, where $\Omega \subset \Lambda(A)$.

Algorithm 1 Modified Lanczos method in floating point arithmetic.

Input: Symmetric matrix $A \in \mathbb{R}^{N \times N}$, vector $b \in R^N$, $M \in \mathbb{N}$, function g, tolerance τ.
Output: Basis matrix $V_l = [v_1, \dots, v_l] \in R^{N \times l}$ for $K_l(A, b)$, vectors $\alpha = \{\alpha_1, \dots, \alpha_l\} \in \mathbb{R}^l$, $\beta = [\beta_1, \dots, \beta_{l-1}] \in \mathbb{R}^{l-1}$, where $l \in \{1, \dots, M\}$ is the stopping iteration index.
Initialise $V \leftarrow \mathbf{0}^{N \times M}$, $\alpha \leftarrow \mathbf{0}^N$, and $\beta \leftarrow \mathbf{0}^{N-1}$
$v_1 \leftarrow \frac{b}{\|b\|}$
for $j = 1, \dots, M - 1$ **do**
    $w \leftarrow Av_j$
    $\alpha_j \leftarrow v_j^\top w$
    if Partial reorthogonalization **then**
        $w \leftarrow w - \alpha_j v_j$
        if $j > 0$ **then** $w \leftarrow w - \beta_{j-1} v_{j-1}$ **end if**
    else if Full reorthogonalization **then**
        $w = w - \sum_{i=1}^{j} (w^\top v_i) v_i$
        $w = w - \sum_{i=1}^{j} (w^\top v_i) v_i$
    end if
    $\beta_j = \|w\|_2$
    if $\beta_j = 0$ **then**
        return $V_j, \alpha_{1:j}, \beta_{1:j-1}$ ▷ — Breakdown —
    else
        if $\|g_j - g_{j-3}\|_2 < \tau$ **then**
            return V_j, α, β ▷ — Breakdown —
        end if
        $v_{j+1} = \frac{w}{\beta_j}$
    end if
end for
return $V_M, \alpha_{1:M}, \beta_{1:M-1}$.

Algorithm 2 Approximation of $g(\mathcal{L})s$ using Lanczos.

Input: Function g, Matrix T, signal s.
$\lambda, U \leftarrow$ diagonalize(T) ▷ Here λ is the vector of eigenvalues.
$u_1 \leftarrow (U^\top)_{:,1}$
$y \leftarrow \|s\| U(\text{Diag}(g(\lambda))U^\top)e_1 = \|s\| U(g(\lambda) \odot u_1)$
▷ Here $u \odot v$ denotes the element-wise product between two vectors u and v.
return Vy

Algorithm 3 Evaluation of $g(\mathcal{L})s$ using Fourier basis.

Input: Function g, Graph $\mathcal{G}$, signal s.
$\Lambda, U \leftarrow$ Fourier_basis($\mathcal{G}$)
return $Ug(\Lambda)U^\top s$

Algorithm 4 Orthogonalization selection.

Input: $\delta = 10^{-10}$.
$V, T, j \leftarrow$ Lanczos(A,b, Partial)
if $\|V^\top V - I\|_2 > \delta$ **then**
    $V, T, j \leftarrow$ Lanczos(A,b, Full)
end if

1.3.1 Computational cost

Here we provide an overview of the computational cost of the algorithms we tested, for an input sparse matrix A, where nnz(A) is the number of non-zero elements of A:

- **Lanczos:** Computes V with partial orthogonalization, with $O(MN + M \cdot \text{nnz}(A))$. Note that the product Av_j dominates the cost, and that nnz$(A) = N^2$ when A is dense. When we need to evaluate function g at each iteration, such as in Section 2.2, we must consider also the $O(N^2)$ operations to compute $g(\mathcal{L})$ with divide and conquer, thus the total cost in this case becomes $O(MN^2)$.
- **Lanczos with double orthogonalization:** It differs from Lanczos by the computation of $w = w - \sum_{i=1}^{j} v_i(v_i^\top w)$, which can cost up to $O(MN)$. Thus the total cost becomes $O(M^2N + M \cdot \text{nnz}(A))$, and when we also evaluate g it becomes $O(M^2N + MN^2)$.
- **Arnoldi with double orthogonalization**: For the orthogonalization it takes $O(M^2N)$, plus the matrix vector product at each step with A, hence again $O(M^2N + M \cdot \text{nnz(A)})$. Cost is higher when evaluating g, due to the fact that matrix H is not perfectly symmetric (it is up to a small epsilon) and not tridiagonal (it has some small perturbations outside the diagonals). Hence we must use the Schur-Parlett algorithm to compute the matrix function (which we use in the routine `scipy.linalg.funm`), which adds $O(N^3)$ complexity per iteration, and in total $O(MN^3)$ in the worst case.

We observe that in Lanczos computing g it not necessarily a drawback when A is dense or when few iterations are performed. It should also be noted that these algorithms suffer from a significant $O(MN)$ memory cost for storing V_M.

1.3.2 Motivating double orthogonalization

From the experimental setting in Section 2.1, we observed that the orthogonality of the basis V with Arnoldi/Lanczos with a single orthogonalization step was worse than with standard Lanczos, while double orthogonalization showed good results. One reason could be that the Laplacian matrices that we are considering are badly conditioned. The following example motivates this choice:

Example 1.11. We consider a Erdős-Rényi graph with $N = 1000$, $M = 200$, $p = 0.04$. The associated Laplacian $\mathcal{L}$ is terribly conditioned, indeed $k_2(\mathcal{L}) = 2.365029616834158 \times 10^{17}$:

As we can observe in Table 1, orthogonalizing twice against the previous basis vectors yields a better orthogonal basis V, while a single Gram-Schmidt iteration performs worse than standard Lanczos.

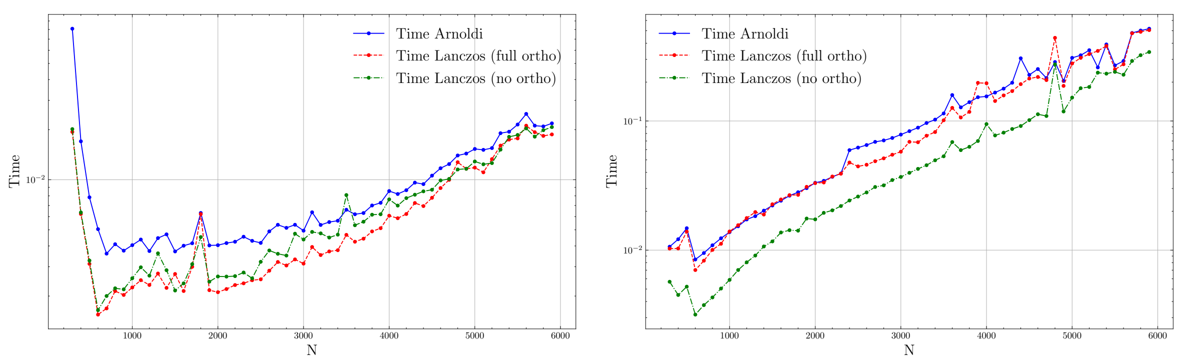
Table 1: Correlation between orthogonalization type and basis orthogonality for Lanczos.

Orthogonalization type	$\|V^\top V - I\|_2$
Partial	$6.7883747792312356 \times 10^{-15}$
Single Gram-Schmidt orthogonalization	$1.7133096895857787 \times 10^{2}$
Double Gram-Schmidt orthogonalization	$6.7883747792312356 \times 10^{-15}$

1.3.3 Comparing Lanczos with Arnoldi

Here we do a short numerical comparison of Lanczos and Arnoldi with double orthogonalization, which should match the cost analysis from Section 1.3.1. We do this to justify the need for using Lanczos with double orthogonalization rather that Arnoldi. In Algorithm 5 we report our implementation of Arnoldi.

We consider the function g used in the experiments and $N \in \{300, 400, 500, \ldots, 6000\}$, $M = 200$ and $p = 0.04$. In Figure 2 and Table 2, we observe different correlations when running the algorithm with residual break and until M. In the former, described in Figure 2a, Lanczos with full orthogonalization outperforms the other two. In the latter in Figure 2b, curves generally match theoretical results, despite some small cases of the full-orthogonalization variant. Looking better at Figure 2a, we see that full orthogonalization is converging faster than predicted in the complexity analysis due some eigenvalues of T_l being evaluated at zero by g, as we will observe better soon in the experiments in Section 2. Also Arnoldi is slower than the other two algorithms, which confirms the increased complexity due to Schur-Parlett method for evaluating the function.



(a) Algorithms stop when $\|g_{l-3} - g_l\|_2 < \tau = 10^{-6}$.

(b) Algorithms run until last index M.

Figure 2: Comparing Lanczos and Arnoldi.

2 Experiments

In the following section we generate random Erdős-Rényi and Sensor graphs, as defined in Definitions 1.4 and 1.5, to study the Lanczos algorithm's accuracy and performance.

Algorithm 5 Arnoldi with double orthogonalization.

Input: Matrix $A \in R^{N\times N}$, $s \in \mathbb{R}^N$, $M \in \mathbb{N}$, function g, tolerance τ.
Output: Basis matrix $V_l = [v_1, \dots, v_l] \in R^{N\times l}$ for $K_l(A, b)$, upper-Hessenberg matrix $H \in R^{l\times l}$ where $l \in \{1, \dots, M\}$ is the stopping iteration index.
$v_1 \leftarrow s/\|s\|_2$
Initialise $V \leftarrow \mathbf{0}^{N\times M}$ and $H \leftarrow \mathbf{0}^{M\times M}$
$V_{:,1} \leftarrow v_1$
for $j = 1 \dots M-1$ **do**
    $w_1 \leftarrow Av_j$
    ▷ First orthogonalization
    $h_1 \leftarrow V_j^\top w_1$
    $w_1 \leftarrow w_1 - V_j \cdot h_1$
    if $w_1 = 0$ **then break end if**
    $v_{j+1} \leftarrow w_1/\|w_1\|_2$
    ▷ Second orthogonalization pass
    $h_2 \leftarrow V_j^\top v_{j+1}$
    $w_2 \leftarrow v_{j+1} - V_j \cdot h_2$
    $H_{1:j,j} \leftarrow h_1 + h_2$
    $H_{j+1,j} \leftarrow \|w_1\|_2\|w_2\|_2$

    if $H_{j+1,j} = 0$ **then**
        break ▷ Arnoldi breakdown
    end if
    ▷ Convergence check
    if $j > 3$ **and** $j < M$ **then**
        if $\|g_j - g_{j-3}\|_2 < \tau$ **then**
            return $V_j, H_{1:j,1:j}$
        end if
    end if
    $v_{j+1} \leftarrow w_2/\|w_2|_2$
    $V_{:,j+1} \leftarrow v_{j+1}$
end for
return $V_{:,1:j}, H_{1:j,1:j}$

2.1 Comparing global error with iterate difference

We want to numerically verify the approximation in Equation (1) by relating the residuals in Definition 1.9. Let's recall some quantities that will be useful in this section: N is the number of graph nodes and size of the Laplacian $\mathcal{L}$, M is the maximum size of the Krylov subspace, and p the probability of the graph.

Experiment. We set $(N, M, p) = (500, 200, 0.04)$ and we try to replicate the Experiment 1 in Article [3].

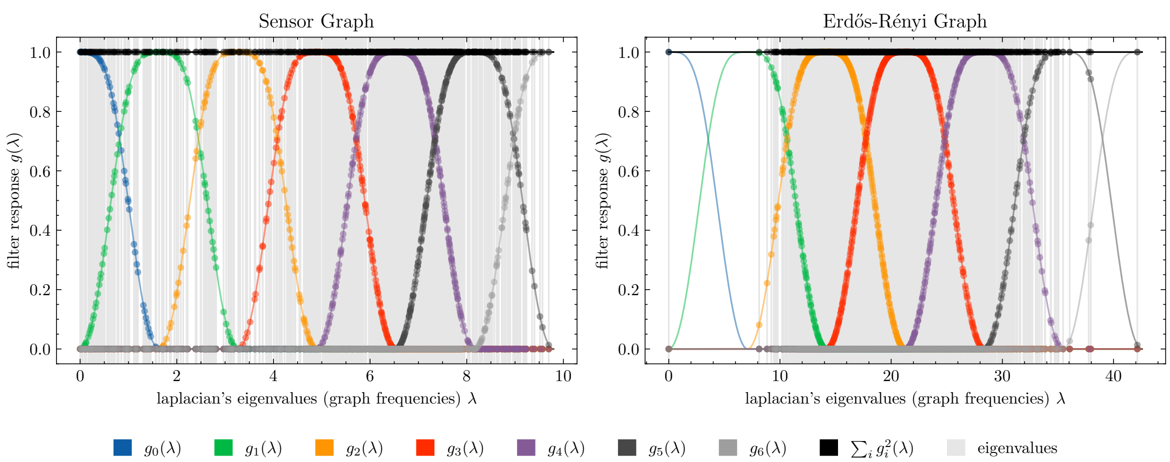
Recall that computing $g(\mathcal{L})s$ is done by computing the Fourier basis for the graph $\mathcal{G}$, which is feasible for the small graph that we are considering.

Table 2: Mean and variance of values in Figure 2.

	With τ bound		Run until M	
	Mean	Variance	Mean	Variance
Time Lanczos (no ortho)	0.007253	0.000032	0.078663	0.008244
Time Lanczos (full ortho)	0.006678	0.000032	0.131559	0.019279
Time Arnoldi	0.010095	0.000125	0.137977	0.018646
Stopping index Lanczos (no ortho)	11.614035	25.812657	200	0
Stopping index Lanczos (full ortho)	11.684211	31.005639	200	0
Stopping index Arnoldi	11.684211	31.005639	200	0

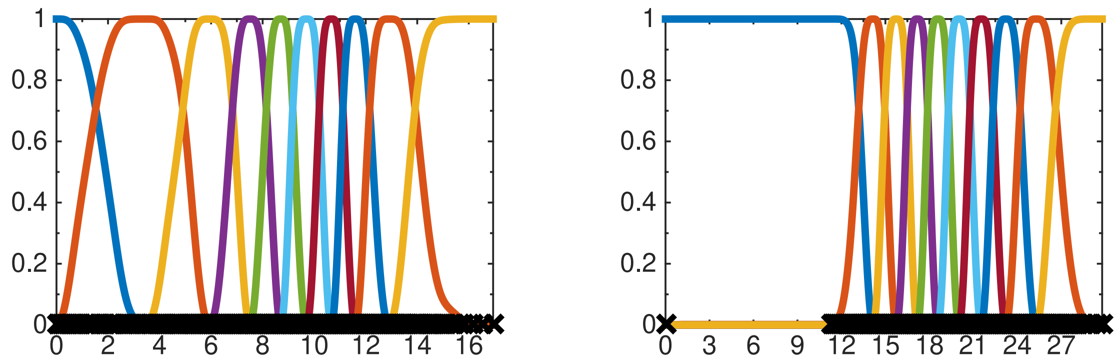
We consider the function $g(t) = \sin(\frac{1}{2}\pi\cos^2(\pi t))\chi_{[-\frac{1}{2},\frac{1}{2}]}$, which we adapt to the graph spectrum inteval $[0, \lambda_{\max}]$, and it is implmented in the `Itersine` class in Pygsp in Figure 3. Unfortunately, the authors in [3] apply an additional transformation to g to match the eigenvalue distribution of $\mathcal{L}$, which is showed in 4. We were not able to reproduce it (despite many trials) as it seemed not documented both in the article and library. Without this adaptation, when we evaluate the eigenvalues of T_l using g, some eigenvalues vanish, especially in the case of Erdős-Rényi graphs. Using the algorithm described in Section 1.3, and computing the errors $\|e_l\|_2$ and $\|g_{l+3} - g_l\|_2$ for every $l = 1, \dots, M$, we obtain the plot in Figure 5.

Figure 3: Itersine plot.



Looking closely at Figure 5: we immediately observe that the error curves are close, and that their value is decreasing, which shows that the approximation of g gains precision as the basis V_l gets larger. We also see how the eigenvalues of $\mathcal{L}$ and T influence the steepness and smoothness of the curves: when eigenvalues fall outside $[-\frac{1}{2}, \frac{1}{2}]$, we see that the curves rapidly reach machine precision: this is unfortunately caused by issue explained above. We also observe evident lack of smoothness in the iteration error curve: we believe that since $g(t) = 0$ for $t \in \mathbb{R} \setminus [-\frac{1}{2}, \frac{1}{2}]$, the eigenvalues of T tend to jump in and out of the interval $[-\frac{1}{2}, \frac{1}{2}]$, causing spikes. Hence, we found that this method is sensitive to the eigenvalues distribution of the Laplacian.

We also found interesting to visualize graphically as in Figure 1 how the signal s is filtered by the algorithm on



(a) Itersine of sensor graph. (b) Itersine of Erdős-Rényi graph.

Figure 4: Itersine plot reported in [3].

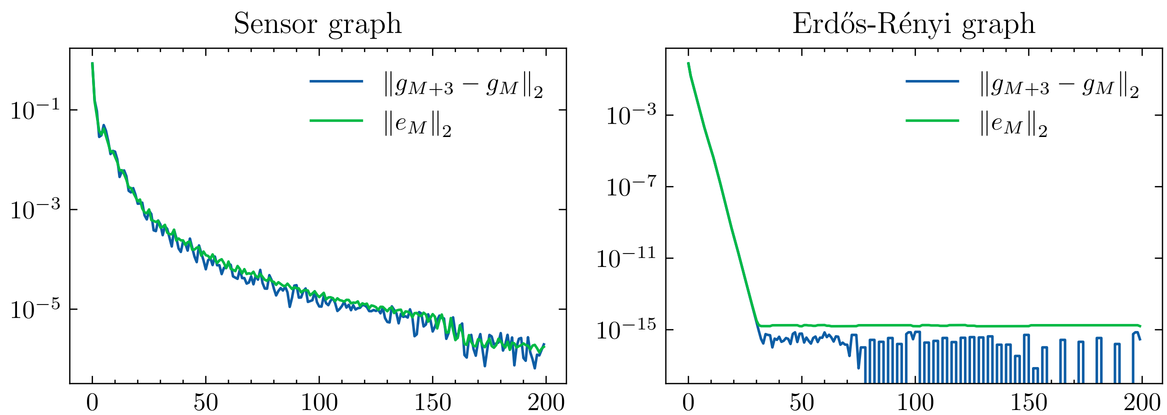


Figure 5: Comparing iteration error with true error.

the graphs used in this experiment, which corresponds to applying a low-pass filter.

2.2 Performance comparison for Erdős-Rényi graphs

In this experiment, we will study the performance correlation for running Lanczos on Erdős-Rényi graphs $\mathcal{G}(N, p)$ and comparing execution time when only N varies and when only p varies.

Remark 2.1 (Role of N and p). Erdős-Rényi graphs, the parameters N and p determine dimension and sparsity of $\mathcal{L}$, respectively. In fact, when N increases, we have a larger Laplacian and we expect higher computational cost due to the increased cost of the matrix-vector multiplication $\mathcal{L}v$ and to compute the eigenvalues of T_M. When p increases, $\mathcal{L}$ becomes more dense, requiring more floating point operations for the algorithm and so increasing runtime. Figure 6 shows how this property in a sample of the matrices.

We choose $\delta = 10^{-5}$, as we experimentally observed that smaller thresholds rarely trigger full orthogonalization. As in Algorithm 1 our stopping criteria happens when $\|g_l - g_{l-3}\|_2 < \tau$, at the step l. [2]

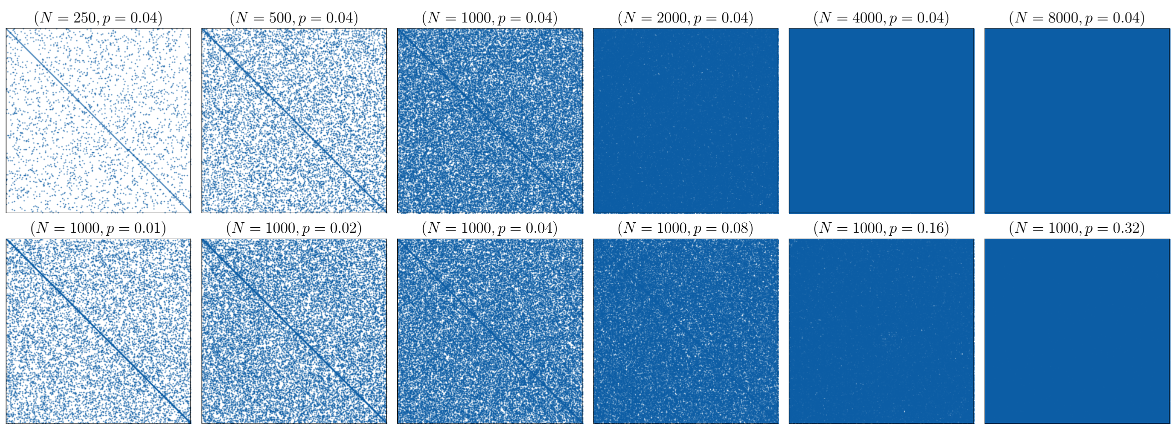
[2]Here index goes backwards because matrix T_j would not be defined for $j > l$ at iteration l!

Experiment. In the following experiment, we generate two sample datasets, letting $n = 6$,:

1. N varies in $\{250, 500, \dots, 2^n \cdot 250\}$, $\tilde{p} = 0.04$.
2. p varies in $\{0.01, 0.02, \dots, 2^n \cdot 0.01\}$, $\tilde{N} = 1000$.

By abuse of notation we will often refer to the tuple (N, p) to indicate the columns of the datasets above as we are interested in correlating increasing couples of $(N, p) \in \{(250, 0.01), (500, 0.02), \dots, (8000, 0.32)\}$. We repeat the generation of these graphs 200 times to avoid bias from graph randomness.

Figure 6: Plot of the sparsity pattern of the generated matrices, as N, p varies.



For time telemetry, we adopt Python's `time.perf_monitor()` which ensures precise measurement of the execution in Algorithm 4. We acknowledge that this method requires running Lanczos twice, but from what we saw for some cases in Table 3 sometimes it is convenient trying standard Lanczos first. From these first examples, we think it's difficult to predict whether the basis will need or not to be reorthogonalized.

We now show plot of some runs of this experiment for some conditions on τ, δ and the stopping criteria.

2.2.1 Run with selective orthogonalization, $\tau = 10^{-5}$ and $\delta = 10^{-5}$

Looking at Table 4 and 7a, we observe that time is higher for $(N, p) = (250, 0.01)$, and then increases gradually. In the case of N, we know from the theory that when $N \gg M$, the Krylov subspace is approximated better by the algorithm, hence time start growing stably from $N \geq 500$, and we see a significant 4x gap from $N = 4000$ and $N = 8000$. When p varies the matrices become dense and we expect longer computational times, as the matrix may lose orthogonality for dense matrices, but we see the opposite in Table 4. Indeed we see time for $p = 0.01$ is more than a tenth of second, which is higher than the other cases, and then oscillates for other values of p, where the τ bound stops the algorithm early. We believe that this is due to the phenomenon we already observed in Section 2.1: the error $g_{l-3} - g_l$ reaches machine precision fast as depicted in Figure 5, because the eigenvalues of the input matrix $\mathcal{L}$, which becomes dense as p grows, are mapped to zero by g.

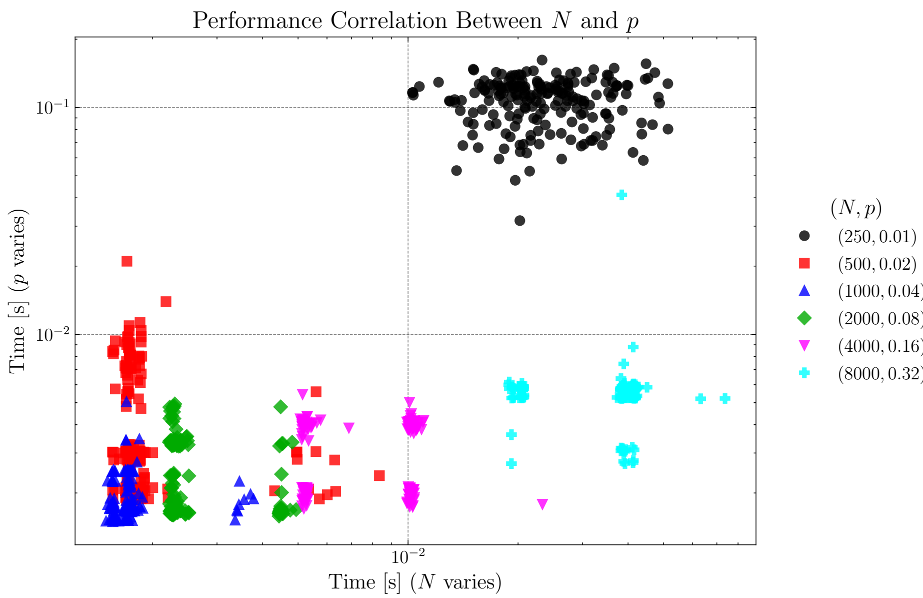
In general Figure 7a shows a linear growth with few outliers, where times for increasing N result slower than those of g. Hence for this stopping criteria the size of $\mathcal{L}$ influences execution time more than the sparsity. We believe that we could see a similar pattern in case the algorithm runs with a different g, as the added complexity is due the evaluation procedure. In the next section we try to run the procedure until the index M, to verify how the error bound is influencing performance.

Table 3: Timing the 12 matrices in Figure 6 for just one run, where N varies and p fixed (top), and vice-versa (bottom).

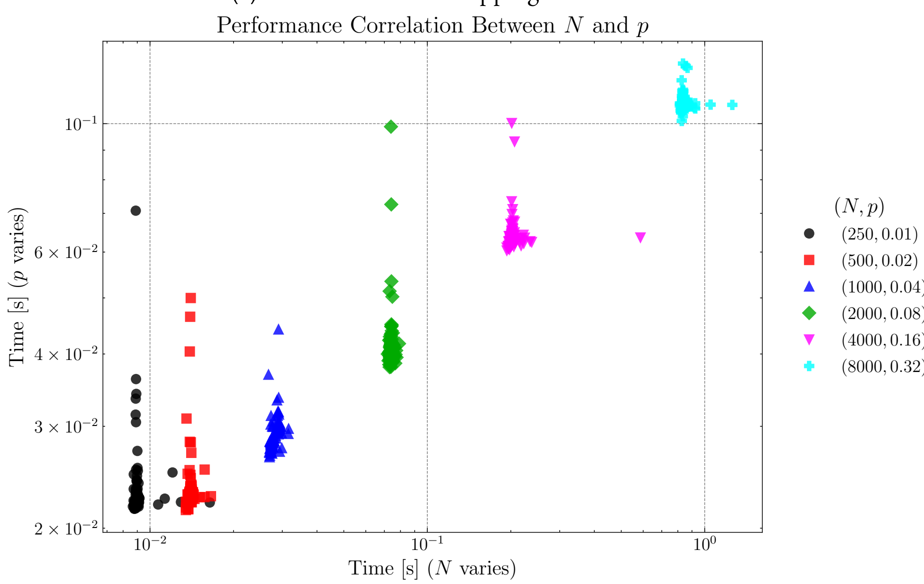
N	M	p	Time (s)	Orthogonalization	Breakdown	Stopping index	ε
250	200	0.04	0.025940	Full	Bound	61	10^{-5}
500	200	0.04	0.002009	Partial	Bound	16	10^{-5}
1000	200	0.04	0.001946	Partial	Bound	13	10^{-5}
2000	200	0.04	0.005585	Full	Bound	11	10^{-5}
4000	200	0.04	0.012243	Full	Bound	10	10^{-5}
8000	200	0.04	0.059831	Full	Bound	9	10^{-5}
1000	200	0.01	0.137948	Full	Bound	131	10^{-5}
1000	200	0.02	0.006453	Full	Bound	23	10^{-5}
1000	200	0.04	0.001641	Partial	Bound	13	10^{-5}
1000	200	0.08	0.003160	Full	Bound	11	10^{-5}
1000	200	0.16	0.003718	Full	Bound	10	10^{-5}
1000	200	0.32	0.002654	Partial	Bound	9	10^{-5}

Table 4: Stats when N and p varies (top and bottom) with iteration bound.

		Time		Stopping index	
N	p	mean	var	mean	var
250	0.04	0.024815	0.000077	61.125000	138.602387
500	0.04	0.001995	0.000001	15.655000	3.031131
1000	0.04	0.001761	0.000000	12.780000	0.212663
2000	0.04	0.002632	0.000001	11.050000	0.047739
4000	0.04	0.008299	0.000007	10.000000	0.000000
8000	0.04	0.036846	0.000070	9.000000	0.000000
p	N				
0.01	1000	0.107721	0.000594	117.410000	111.408945
0.02	1000	0.004034	0.000008	17.865000	14.036960
0.04	1000	0.001913	0.000000	12.770000	0.218191
0.08	1000	0.002411	0.000001	11.000000	0.000000
0.16	1000	0.003342	0.000001	9.840000	0.135075
0.32	1000	0.005640	0.000007	9.000000	0.000000



(a) Correlation when stopping iteration bound.



(b) Correlation when running until M.

Figure 7: Log-log plot of 200 iterations of the comparison experiment.

2.2.2 Run with selective orthogonalization, no error bounds and $\delta = 10^{-5}$

In Figure 7b and Table 5 we observe an almost-linear correlation, with few outliers—these could be caused by the fact that full orthogonalization is triggered in some cases, causing two runs of the algorithm.

Table 5: Stats when N and p varies (top and bottom) without bound.

		Time		Stopping index	
		mean	var	mean	var
N	p				
250	0.04	0.009062	0.000001	200	0
500	0.04	0.014043	0.000000	200	0
1000	0.04	0.028319	0.000001	200	0
2000	0.04	0.074183	0.000001	200	0
4000	0.04	0.204908	0.000764	200	0
8000	0.04	0.841120	0.001346	200	0
p	N				
0.01	1000	0.022907	0.000015	200	0
0.02	1000	0.023276	0.000009	200	0
0.04	1000	0.029179	0.000003	200	0
0.08	1000	0.041019	0.000025	200	0
0.16	1000	0.063665	0.000014	200	0
0.32	1000	0.108326	0.000008	200	0

3 Conclusions & future work

To conclude, we observed how bound and orthogonalization techniques can improve performance trough early breaks, especially for large and dense matrices. However, without the precise rescaling of g, these bounds can start to be unreliable and terminate the algorithm early, which may provide an insufficient approximation of the Krylov basis. We didn't address how the high memory cost of $O(NM)$ of Lanczos could be avoided by using a two-step implementation, or how we could use selective reorthogonalization as described in [4] to reduce time complexity, because both are the scope of this work.

References

[1] Ulrike Von Luxburg. A tutorial on spectral clustering. *Statistics and computing*, 17(4):395–416, 2007.

[2] Michaël Defferrard, Lionel Martin, Rodrigo Pena, and Nathanaël Perraudin. Pygsp: Graph signal processing in python.

[3] Ana Susnjara, Nathanael Perraudin, Daniel Kressner, and Pierre Vandergheynst. Accelerated filtering on graphs using lanczos method. 2015.

[4] James W Demmel. *Applied numerical linear algebra*. SIAM, 1997.